MyDoc: Delivering highly available digital healthcare with Google Cloud Platform

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About MyDoc

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Industries: Healthcare

Location: Singapore

Google Cloud Results

- Increased API speed sevenfold compared to previous cloud environment
- Gained ability to deliver updates without downtime of up to two hours each time
- Achieved pricing flexibility to support cost-effective scaling

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AI Platform (https://cloud.google.com/ai-platform/)

Compute Engine (https://cloud.google.com/compute/)

Operations (https://cloud.google.com/products/operations)

Dialogflow Enterprise Edition (https://cloud.google.com/dialogflow-enterprise/)

Founded in November 2012 and headquartered in Singapore, MyDoc (https://www.my-doc.com/) provides a digital platform that enables corporate employees to access healthcare services such as consultations, prescriptions, and medical screenings. “We saw a gap in the market for patient engagement and digital healthcare in Asia-Pacific," Dr. Vas Metupalle, Co-founder, MyDoc, says. "While there were a number of businesses in the United States providing patient engagement services, the market in this region was primarily limited to applications that allowed people to book appointments with medical practitioners."

When establishing MyDoc, Dr. Metupalle and fellow co-founder Dr. Snehal Patel decided the platform should provide a range of patient engagement services, including video health consultations. In addition, to secure support from prospective clients and partners, MyDoc needed to be clinically led. This meant adhering strictly to clinical protocols, licensing requirements, and telehealth regulations.

"From a technology perspective, we needed to capture data so we could filter and manage patients appropriately," Dr. Metupalle adds. "So essentially we needed to create a service comprising patient health records with a light communications layer."

Business growing rapidly

Dr. Metupalle's experience with teleradiology helped MyDoc achieve the interoperability required to add lab and radiology data to patient records. As corporate employees became more comfortable with MyDoc and understood its platform was highly secure, physician-led, and sufficiently credible to secure partnerships with reputable medical providers, the business grew quickly. Its clients now include Asian insurance companies and government health bodies.

MyDoc is undertaking an ambitious expansion plan. The business conducted its first screening in Hong Kong in May 2017, adding the market to its existing operations in Singapore, Malaysia, and Sri Lanka. MyDoc is now targeting Indonesia and plans to operate in India with a joint venture partner based in Bangalore.

While MyDoc is growth-oriented, Dr. Metupalle is mindful of the importance of maintaining clinical quality. The business places considerable importance on training doctors and healthcare professionals in teleconsulting to optimise patient care through the MyDoc platform.

> “Using Google as our primary provider allows us to access all its leading-edge products while retaining the flexibility to use a multi-cloud architecture to support our expansion if required.”

—Mark Ridley, Chief Technology Officer, MyDoc

Scalability and reliability needed

When starting operations, MyDoc adopted a philosophy of using the cloud provider that best met its needs. In early 2017, the business realized it was not in a position to scale using its incumbent provider. In addition, MyDoc was experiencing outages for which it was hard to trace the cause.

Furthermore, the monolithic architecture of its system made releasing new versions difficult. “I joined the company in March 2016, and not long after that we released version 2 of our product, which was our first big release,” Mark Ridley, Chief Technology Officer, MyDoc, says. “It was a massive exercise, involving all team members and all components, that went through the middle of the night. It also entailed two hours of downtime which, for an online healthcare business, could not be repeated in the future.”

Finally, MyDoc’s existing cloud provider did not provide the flexibility that would enable the business to minimize its costs early in its expansion phase.

> “We moved everything over to Google Compute Engine, which gave us the scalability to easily take on more customers from existing or new regions.”
>
> *—Mark Ridley, Chief Technology Officer, MyDoc*

Google chosen as primary provider

Following a review of the cloud services market, MyDoc elected to adopt Google as its primary cloud provider. “Using Google as our primary provider allows us to access all its leading-edge products while retaining the flexibility to use a multi-cloud architecture to support our expansion if required,” Ridley says.

MyDoc has now migrated its core and administration applications, authentication, communication, and static image services to Google Cloud Platform (https://cloud.google.com/) data centres in Singapore and Taiwan.

Containerisation and a microservices architecture

“We moved everything over to Google Compute Engine (https://cloud.google.com/compute/), which gave us the scalability to easily take on more customers from existing or new regions,” Ridley says. “Google also paved the way for us to use containers and to break our system down into an easier-to-manage microservices-based architecture.” In mid-2017, MyDoc formally made the decision to adopt the new architecture and use Kubernetes through Google Kubernetes Engine (https://cloud.google.com/kubernetes-engine/) to deploy, scale, and manage its applications in a containerised environment. The business anticipates completing the move to Kubernetes in Q3 2018, and as of February 2018, was about 40% of the way through its microservices project. “While we were building our own clusters, it made more sense to use a managed product such as Google Kubernetes Engine,” Ridley says.

The business has also added Google Stackdriver (https://cloud.google.com/stackdriver/) to provide error alerts for its in-house-developed login system.

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New release downtime eliminated

MyDoc is already reaping the rewards of its revised technology strategy. “Breaking our system down into microservices allows us to take some services down without affecting others, while moving to Kubernetes means we can perform rolling updates without anyone experiencing any disruption," Ridley says. "So, the two hours or so of downtime associated with new releases is now a thing of the past."

SLA uptime requirement achieved

MyDoc is also meeting its service level agreement that stipulates 99.9% uptime. "We are a 24-hour healthcare service, so it's extremely important that we're available," Ridley says. "The stability of Google, combined with our containerisation and microservices projects, is really helping us." The business has eliminated the outages that affected its operations while running on the previous cloud provider.

A sevenfold API speed improvement

Once MyDoc started replica virtual machine instances in Google Cloud Platform and completed its migration, its API started running seven times faster. This has proven critical to the success of the business. "It is hugely important for us to be able to share and consume external APIs to provide a complete service to our patients," Ridley explains. "By integrating with labs or other third-parties, we can collect and process more data."

The presence of Google Cloud Platform in several regions worldwide supports MyDoc's strategy of deploying Kubernetes clusters close to groups of users to minimize latency. "Our plan is to have several replicas of our APIs and systems positioned in a range of local markets," Ridley says. Storing data in the Google Cloud Platform region in Singapore, where the laws and regulations governing telehealth are quite advanced, has also helped MyDoc lock in partnerships with several insurers in the region. “We’ve had audits from several insurers and they have not had any issues with where and how we’ve stored our data,” Dr. Metupalle says.

Machine learning trains chatbot

MyDoc has also turned to newer Google Cloud Platform products such as Google Cloud Dialogflow (https://cloud.google.com/dialogflow-enterprise) to enhance its offering. The business has used the development suite to train a chatbot before launching it into production in mid-February 2018.